

## Questions

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1. Find the exact value of the improper integral

$$\int_0^4 \frac{1}{\sqrt{x(4-x)}} \, dx$$

[5]

**2.** Show that

$$\int_0^3 \frac{1}{x^2 - 2x + 5} dx = \frac{1}{2} \arctan(3) \quad [5]$$

3. (i) Prove, from definitions involving exponentials, that

$$1 - \tanh^2 u = \operatorname{sech}^2 u \quad [3]$$

- (ii) Prove that, for  $|y| < 1$ ,

$$\operatorname{artanh} y = \frac{1}{2} \ln \left( \frac{1+y}{1-y} \right) \quad [4]$$

- (iii) Use the substitution  $x = 4 \tanh u$  to show that, for  $|x| < 4$ ,

$$\int \frac{1}{16 - x^2} dx = \frac{1}{4} \operatorname{artanh} \frac{x}{4} + c$$

Hence deduce that

$$\int \frac{1}{16 - x^2} dx = \frac{1}{8} \ln \left( \frac{4+x}{4-x} \right) + c$$

where  $c$  is an arbitrary constant.

[6]

- (iv) By first expressing  $-t^2 + 6t + 7$  in completed square form, show that

$$\int_1^5 \frac{1}{-t^2 + 6t + 7} dt = \frac{1}{4} \ln 3 \quad [5]$$

4. (i) For  $x > 5$ , find

$$\int \frac{1}{\sqrt{x^2 - 6x + 5}} \, dx$$

expressing your answer first in terms of arcosh and hence in logarithmic form.

[3]

- (ii) Use integration by substitution with  $x - 3 = 2 \cosh u$  to show that, for  $x > 5$ ,

$$\int \sqrt{x^2 - 6x + 5} \, dx = 2 \left( \frac{x-3}{2} \sqrt{\frac{(x-3)^2}{4} - 1} - \ln \left| \frac{x-3}{2} + \sqrt{\frac{(x-3)^2}{4} - 1} \right| \right) + c$$

where  $c$  is an arbitrary constant.

[7]

**5.**

$$g(x) = \frac{2x^2 - 3x + 31}{x^3 - 3x^2 + 9x + 13}$$

(a) Express  $g(x)$  in the form

$$\frac{P}{x+1} + \frac{Q}{x^2 - 4x + 13}$$

where  $P$  and  $Q$  are constants to be found.

**[3]**

(b) Find  $\int g(x) \, dx$ , giving your answer in the form

$$A \ln |x+1| + B \arctan \left( \frac{x-2}{3} \right) + c$$

where  $A$  and  $B$  are constants to be found.

**[4]**

(c) Hence show that  $\int_0^\infty g(x) \, dx$  diverges.

**[2]**

6. Show that

$$\int_1^4 \frac{1}{\sqrt{10+8x-2x^2}} \, dx = \frac{1}{\sqrt{2}} \left( \arcsin\left(\frac{2}{3}\right) + \arcsin\left(\frac{1}{3}\right) \right) \quad [6]$$

7. (a) Explain why

$$\int_0^{\infty} \frac{1}{x^2 - 6x + 18} dx$$

is an improper integral.

[1]

(b) Show that

$$\int_0^{\infty} \frac{1}{x^2 - 6x + 18} dx = k\pi$$

where  $k$  is a constant to be determined.

[4]

8. Evaluate

$$\int_0^4 \frac{1}{x^2 - 4x + 8} \, dx$$

giving your answer as an exact multiple of  $\pi$ .

[8]

**9.**

$$f(x) = \frac{6x + 1}{4x^2 - 8x + 13}$$

(a) Find  $\int f(x) \, dx$ , giving your answer in the form

$$A \ln(4x^2 - 8x + 13) + B \arctan\left(\frac{2x - 2}{3}\right) + c$$

where  $c$  is an arbitrary constant and  $A$  and  $B$  are constants to be found.

**[4]**

(b) Hence show that the mean value of  $f(x)$  over the interval  $[1, \frac{5}{2}]$  is

$$\frac{1}{2} \ln 2 + \frac{7\pi}{36}$$

**[3]**

(c) Hence write down the mean value of  $3 - 2f(x)$  over the interval  $[1, \frac{5}{2}]$ .

**[1]**

**10.**

$$g(x) = \frac{4x - 1}{x^2 - 4x + 8}$$

Show that

$$\int g(x) \, dx = A \arctan\left(\frac{x-2}{2}\right) + B \ln(x^2 - 4x + 8) + c$$

where  $c$  is an arbitrary constant and  $A$  and  $B$  are constants to be found.**[5]**

**11.** You are given that  $y = \operatorname{arcosh} x$ , where  $x \geq 1$ .

(i) Show that

$$\frac{dy}{dx} = \frac{1}{\sqrt{x^2 - 1}} \quad [4]$$

(ii) Show, by integrating the result in part (i), that

$$y = \ln \left( x + \sqrt{x^2 - 1} \right) \quad [4]$$

(iii) Hence show that

$$\int_{15/8}^{5/2} \frac{1}{\sqrt{4x^2 - 9}} dx = \frac{1}{2} \left( \operatorname{arcosh} \frac{5}{3} - \operatorname{arcosh} \frac{5}{4} \right)$$

Express this answer in logarithmic form. [4]

**12.** Determine each of the following integrals.

(i)

$$\int \frac{1}{9x^2 + 12x + 20} \, dx \quad [4]$$

(ii)

$$\int \frac{1}{\sqrt{45 + 12x - 9x^2}} \, dx \quad [4]$$

**13.** Without using a calculator, find

(a)

$$\int_{-1/5}^1 \frac{1}{25x^2 - 20x + 13} \, dx$$

giving your answer as a multiple of  $\pi$

[5]

(b)

$$\int_{-1/5}^1 \frac{1}{\sqrt{25x^2 - 20x + 13}} \, dx$$

giving your answer in the form  $p \ln(q + r\sqrt{2})$ , where  $p$ ,  $q$  and  $r$  are rational numbers to be found.

[7]

14. (i) Given that  $\tanh y = x$ , where  $|x| < 1$ , show that

$$y = \frac{1}{2} \ln \left( \frac{1+x}{1-x} \right)$$

and hence that

$$\operatorname{artanh} x = \frac{1}{2} \ln \left( \frac{1+x}{1-x} \right) \quad [6]$$

- (ii) Use your result in part (i), or otherwise, to find

$$\int_{1/5}^{2/5} \frac{1}{4-9x^2} \, dx$$

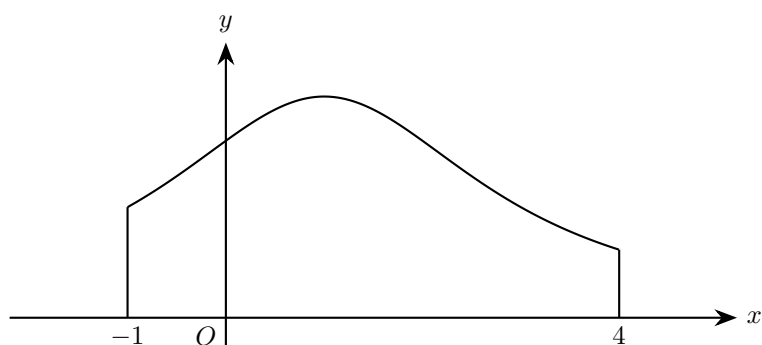
expressing your answer in an exact logarithmic form. [6]

**15.** Show that

$$\int \frac{x^2 + 14x + 21}{(x-1)(x^2 + 4x + 13)} dx = A \ln|x-1| + B \ln(x^2 + 4x + 13) + D \arctan\left(\frac{x+2}{3}\right) + c$$

where  $A$ ,  $B$  and  $D$  are constants to be found.

[7]



- 16.** The diagram shows the cross-section of a solid wooden beam.  
Each unit on the axes represents 6 cm. The curved upper boundary of the beam is modelled by the equation

$$y = \frac{9}{(x - 1)^2 + 4}$$

The cross-section is bounded by the curve, the  $x$ -axis and the lines  $x = -1$  and  $x = 4$ .

Assuming that the beam is a prism of length 240 cm, find, correct to 3 significant figures, the volume of the beam.

[6]

**17.** Find

$$\int_1^e \frac{1}{x\sqrt{4 - (\ln x)^2}} \, dx$$

expressing your answer in terms of  $\pi$ .

[8]